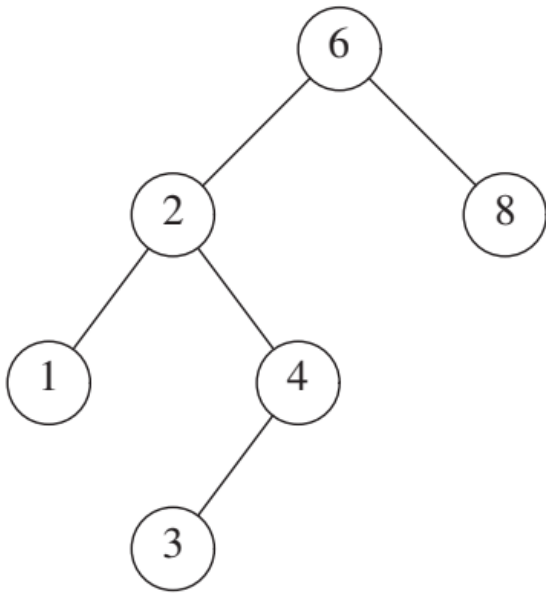


Example for Binary Search Tree: Insert

For example 5.1: if there is **the exist** BinarySearchTree's instance variable name **bst** contains the numbers [1, 2, 3, 4, 6, 8] (In-order traversal) as the image following:



Write the steps for the `_insert_recursive()` function when insert 5 into the instance variable `bst`. In each round, consists of

- 1) what are the values of the parameters `value` and `current_node` of the `_insert_recursive()` function, and
- 2) what statements are executed under the if-else condition that is checked as true.

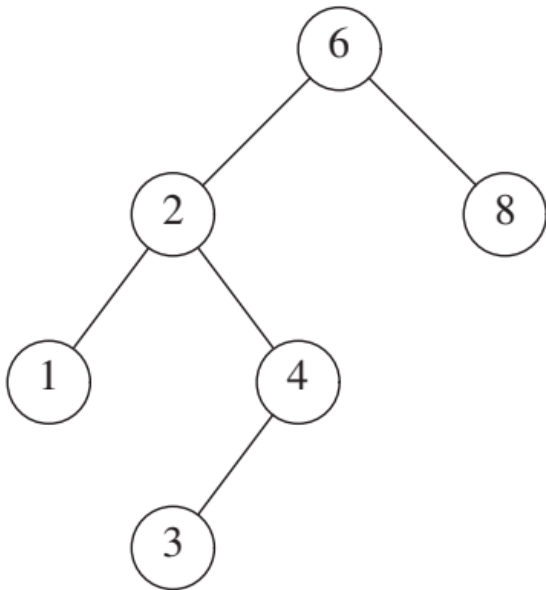
Each iteration must write 2 things.

in main program

```
bst = BinarySearchTree()
```

...

```
bst.insert(5)
```



Answer:

Each iteration for the `_insert_recursive()` function when insert 5

First round value = 5 , current_node = **6**
`_insert_recursive(current_node.left, value)`

round 2: value = 5 , current_node =

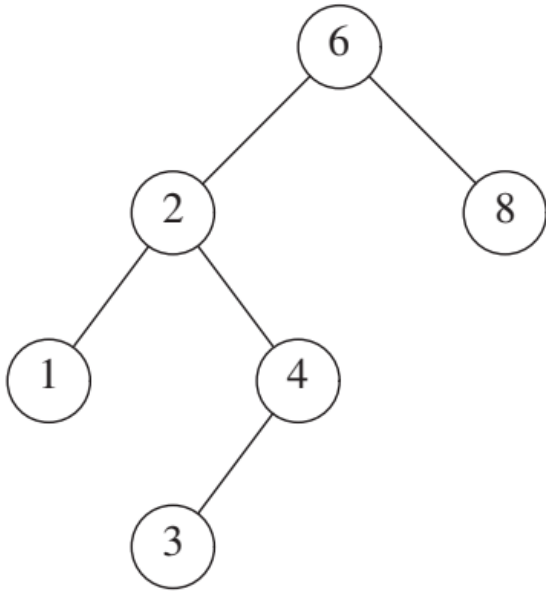
round 3: value = 5 , current_node =



```
round 4: value = 5 , current_node = None  
if current_node.right is None:    is True  
    current_node.right = Node(value)
```

Example for Binary Search Tree: Remove (4), one child.

For example 5.2: if there is **the exist** BinarySearchTree's instance variable name **bst** contains the numbers [1, 2, 3, 4, 6, 8] (In-order traversal) as the image following:

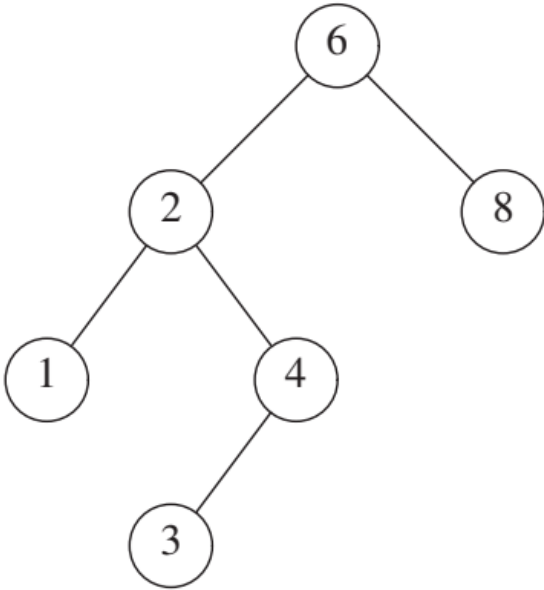


Write the steps for the `_delete_recursive()` function when remove 4 from the instance variable `bst`. In each round, consists of

- 1) what are the values of the parameters `value` and `current_node` of the `_delete_recursive()` function, and
- 2) what statements are executed under the if-else condition that is checked as true.

Each iteration must write 2 things.

Answer:



Each iteration for the `_delete_recursive()` function when remove 4

First round value = 4 , current_node = **6**
current_node.left =
`_delete_recursive(current_node.left, value)`

round 2: value = 4 , current_node =
current_node.right =

round 3: value = 4 , current_node =
elif current_node.right is None: is True
return current_node.left

2

4